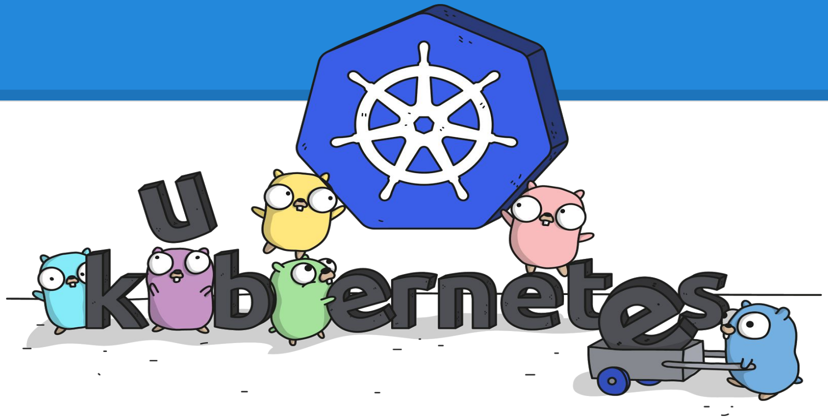
CI/CD for Microservices in Kubernetes



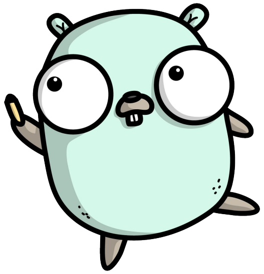
Ananda Dwi Rahmawati – Cloud Engineer
ananda@btech.id

WHO AM I?

- Cloud Engineer at PT Boer Technology (Btech)
- Infrastructur Team at BlankOn Linux Indonesia
- FLOSS Enthusiast
- Mahasiswa Universitas Gadjah Mada
- Tap me at :
  - +62 8132 6789 108
  - t.me/misskecupbung
  - ananda@btech.id
  - https://linkedin.com/in/anandadwir
  - https://misskecupbung.wordpress.com

Documentation

- Kubernetes
  https://kubernetes.io/
- Helm
  https://helm.sh/
- Jenkins
  https://jenkins.io/
- Gitlab
  https://docs.gitlab.com/



What Does “Kubernetes” Mean?

Greek for “pilot” or
“Helmsman of a ship”

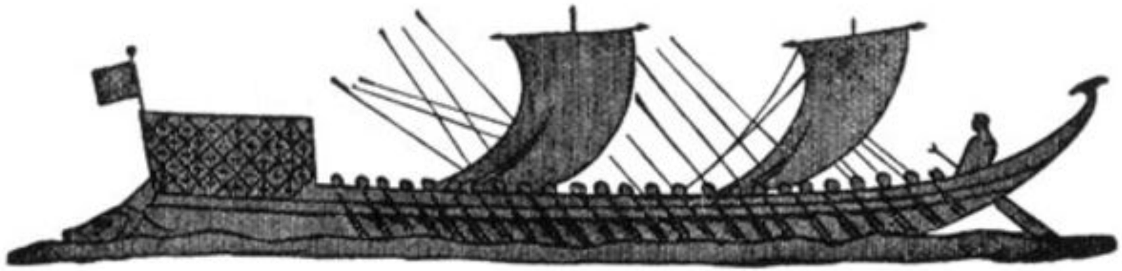
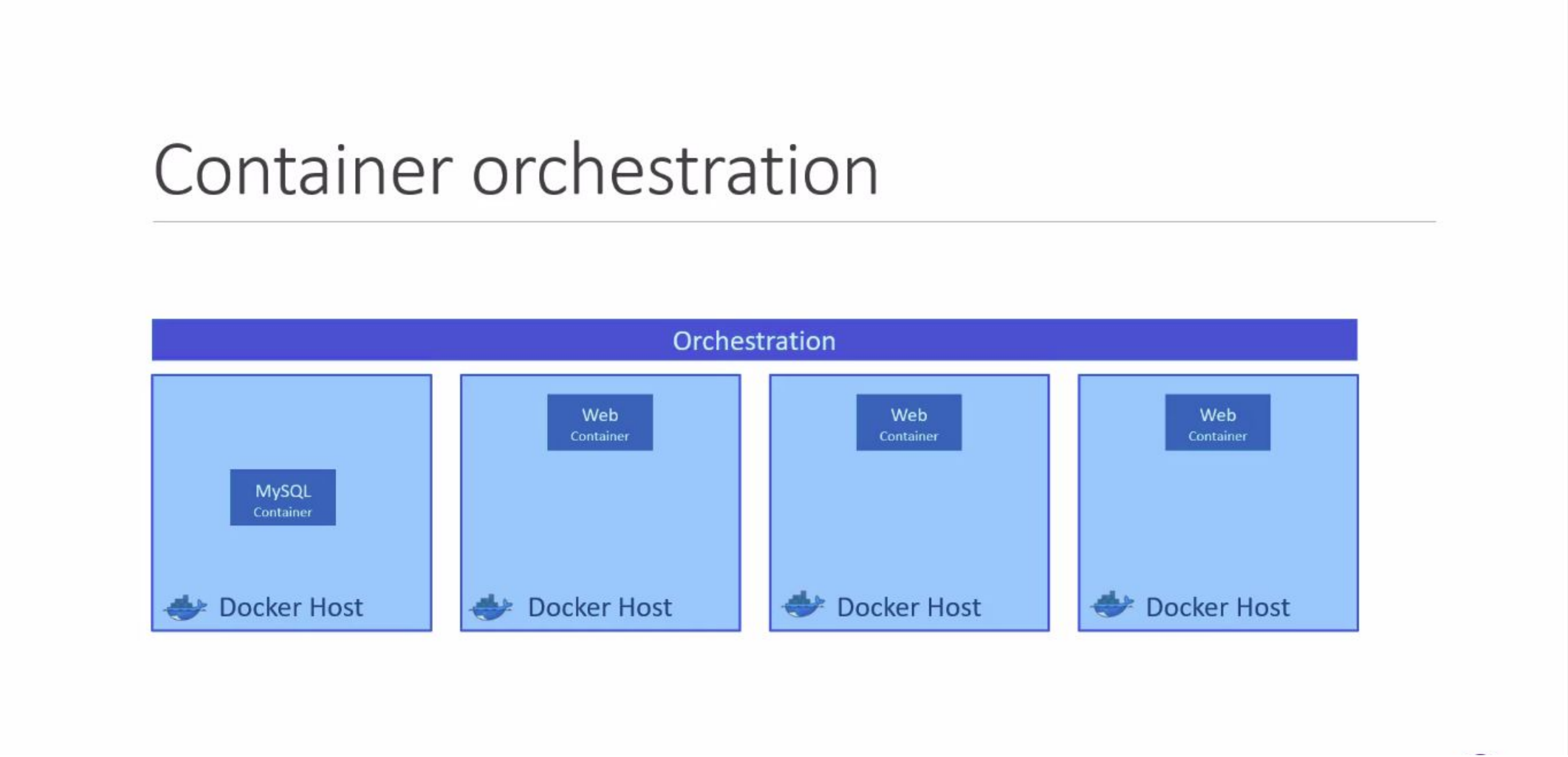


Image Source

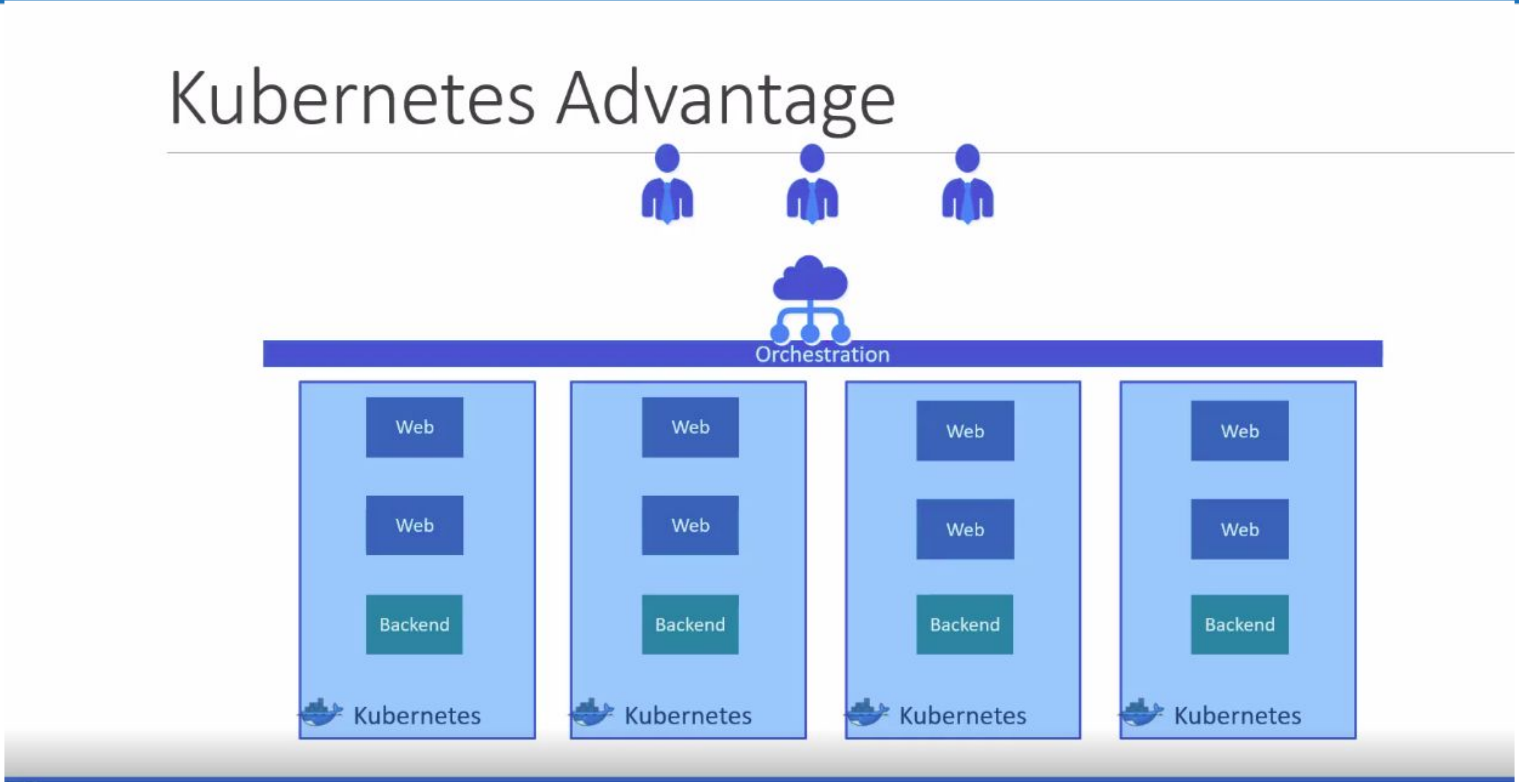
What is Kubernetes?

- Project that was spun out of Google as an open source container orchestration platform.
- Built from the lessons learned in the experiences of developing and running Google’s Borg and Omega.
- Designed from the ground-up as a **loosely coupled** collection of components centered around deploying, maintaining and scaling workloads.

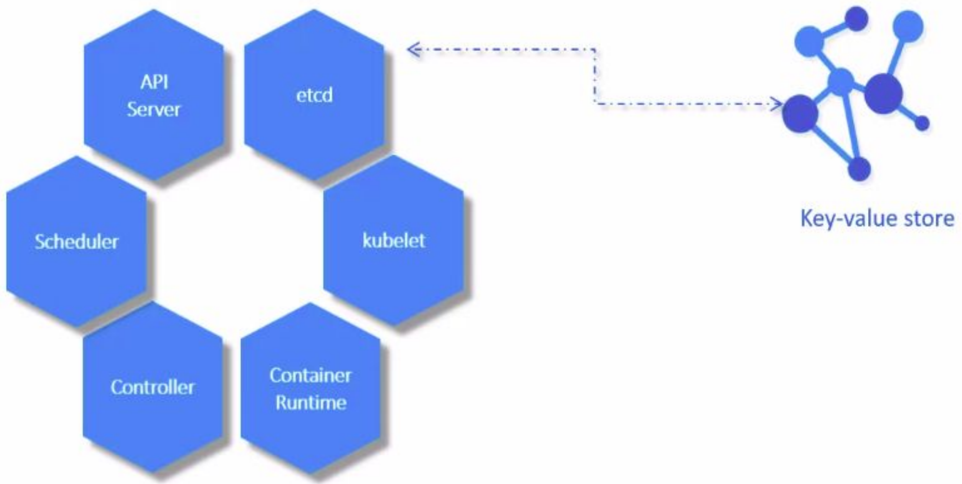
Container Orchestration

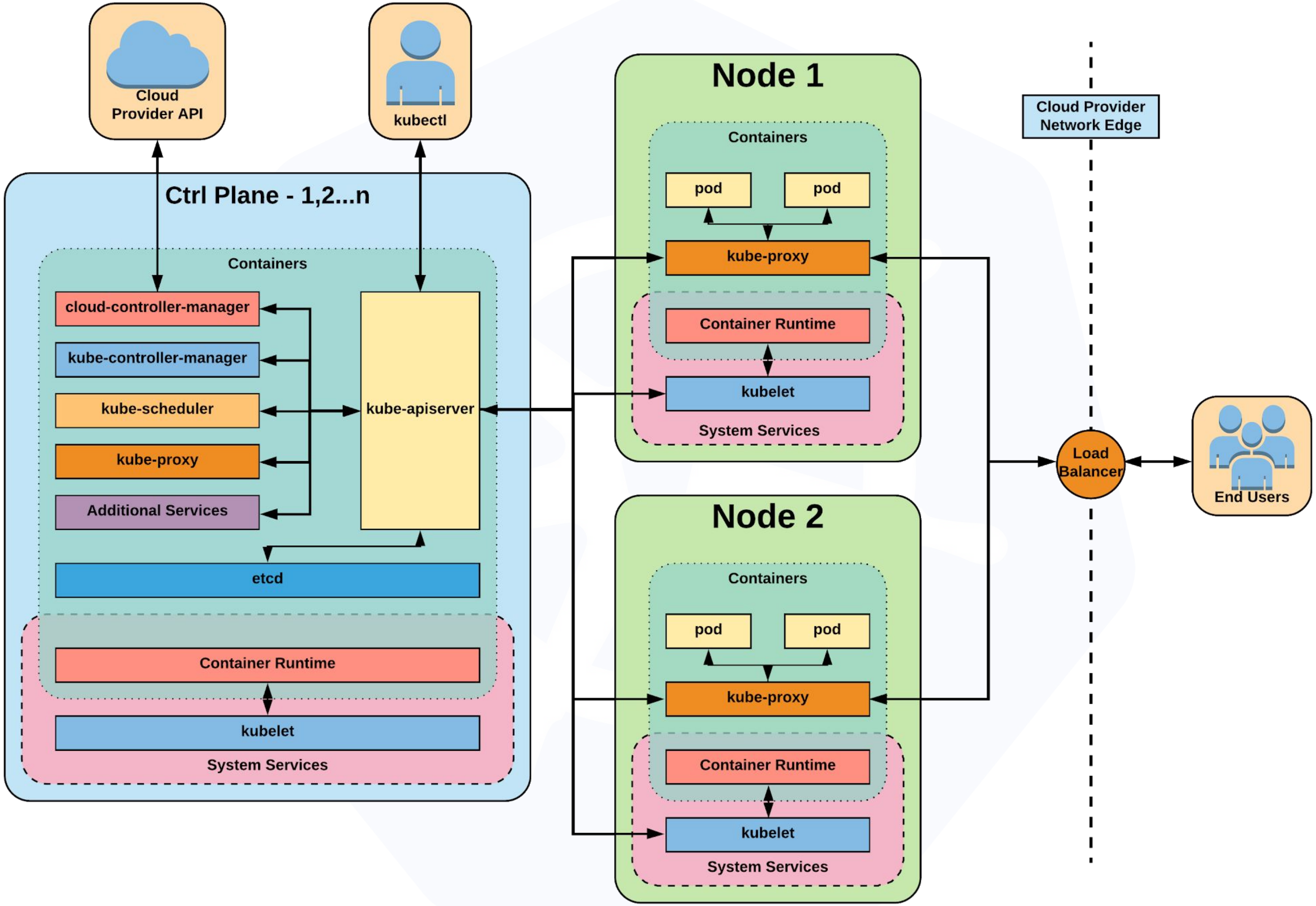


Kubernetes Advantage

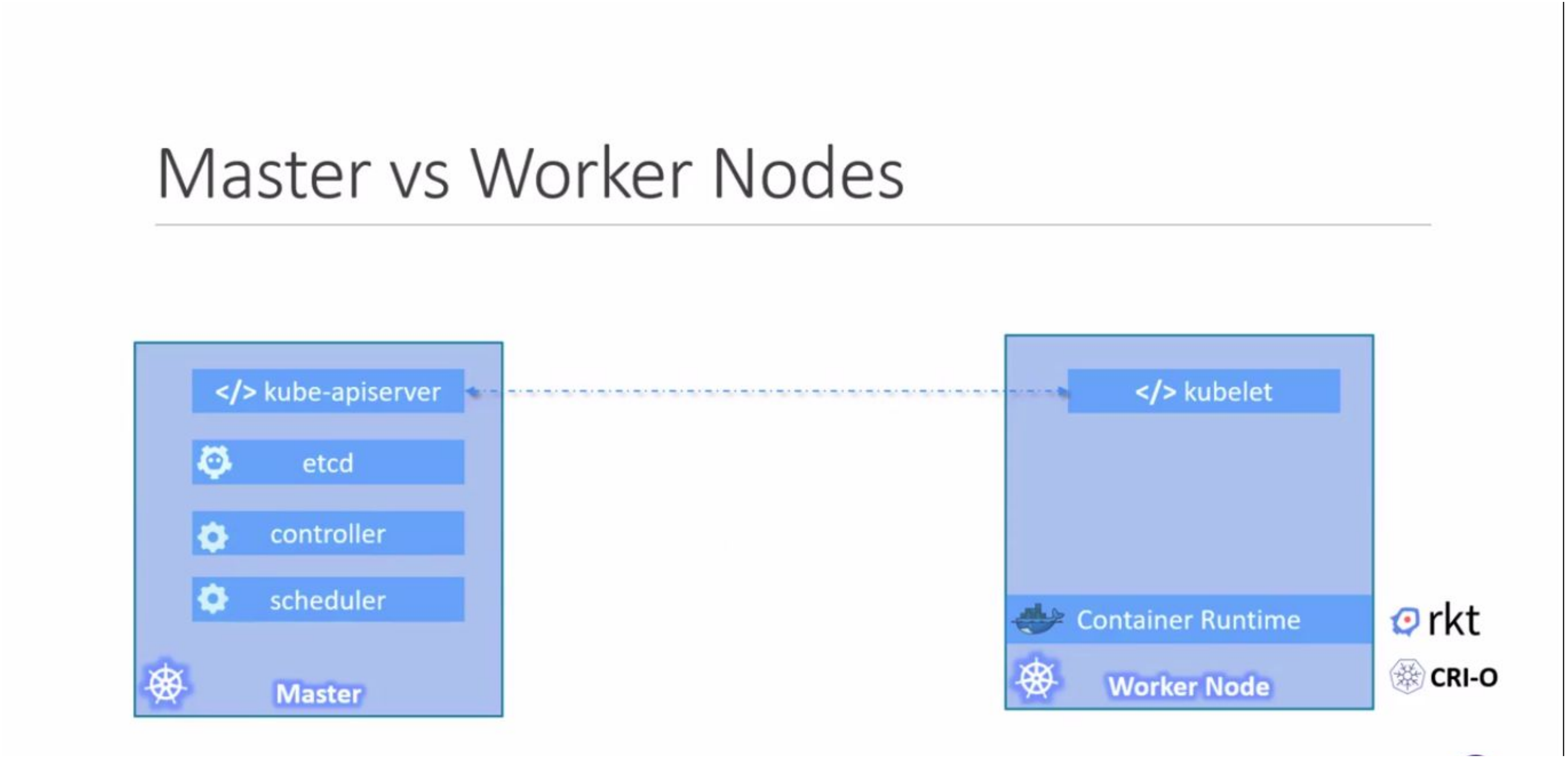


Components

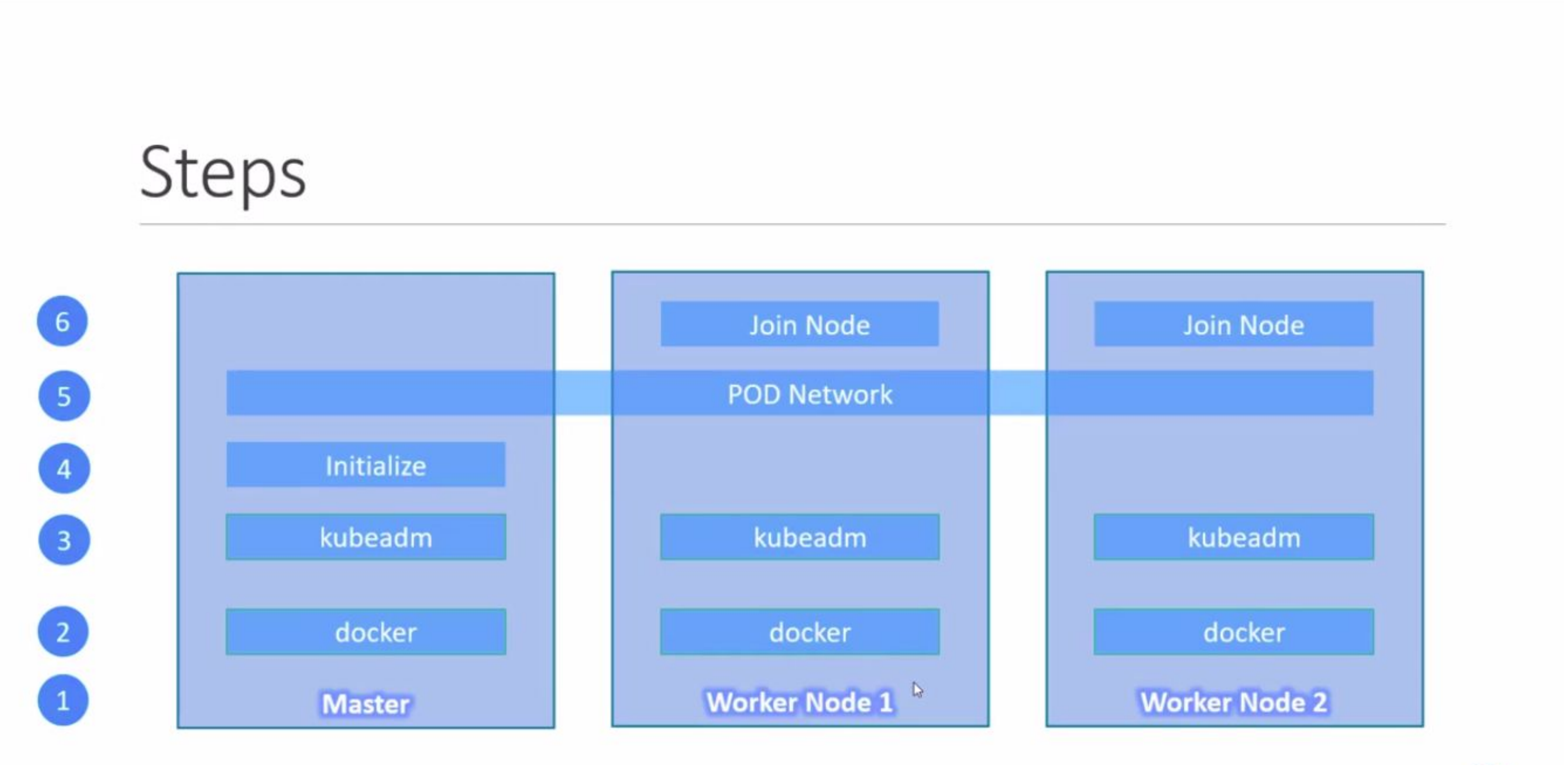

API
Server
etcd
Scheduler
kubelet
Controller
Container
Runtime
Key-value store

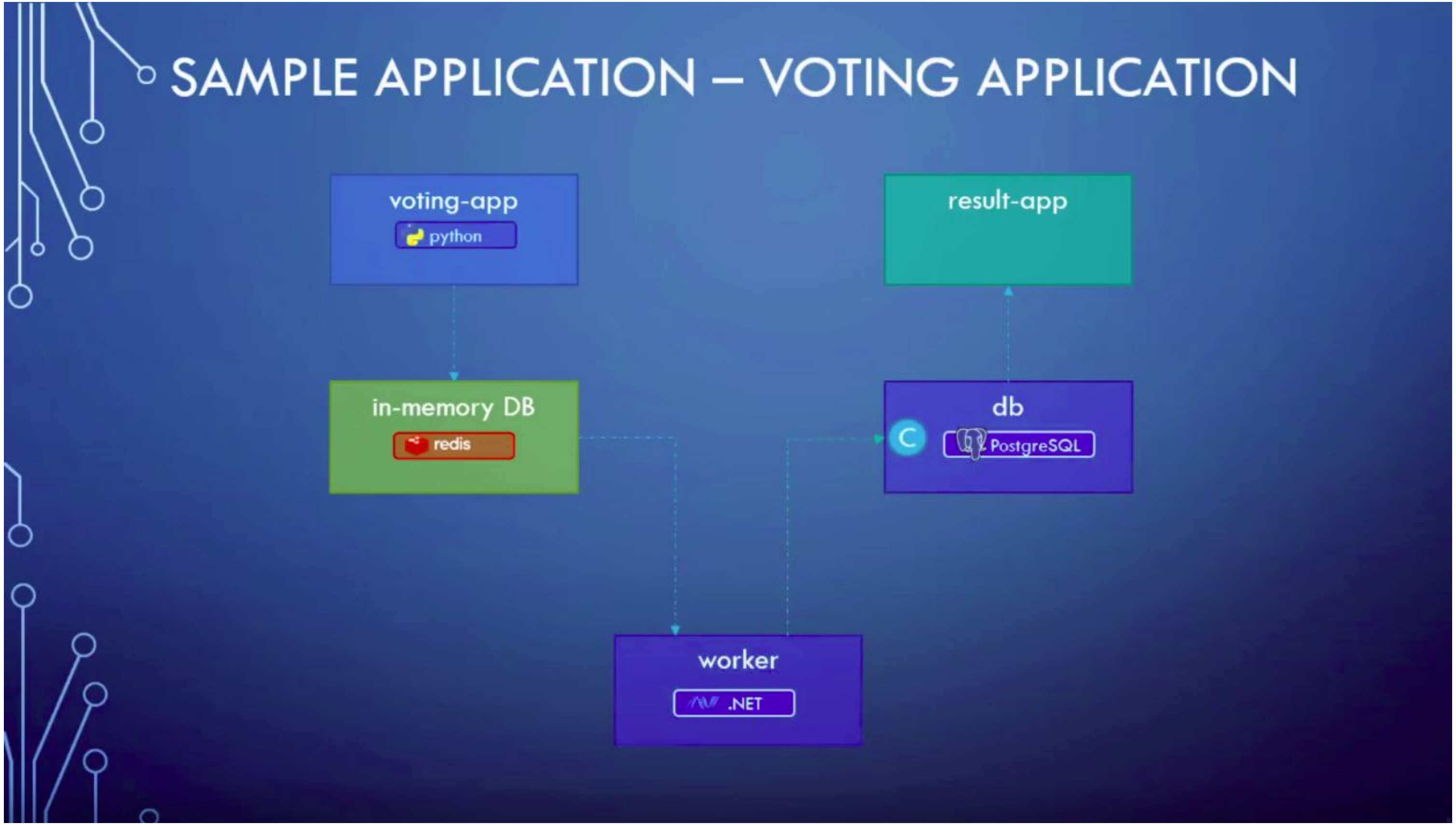

Cloud Provider API
kubectl
Ctrl Plane - 1,2...n
Containers
cloud-controller-manager
kube-controller-manager
kube-scheduler
kube-proxy
Additional Services
kube-apiserver
etcd
Container Runtime
kubelet
System Services
Node 1
Containers
pod
pod
kube-proxy
Container Runtime
kubelet
System Services
Node 2
Containers
pod
pod
kube-proxy
Container Runtime
kubelet
System Services
Cloud Provider Network Edge
Load Balancer
End Users

Kubernetes Component(s)



How To Deploy Simple Cluster




SAMPLE APPLICATION – VOTING APPLICATION
voting-app
python
result-app
in-memory DB
redis
db
C
PostgreSQL
worker
.NET

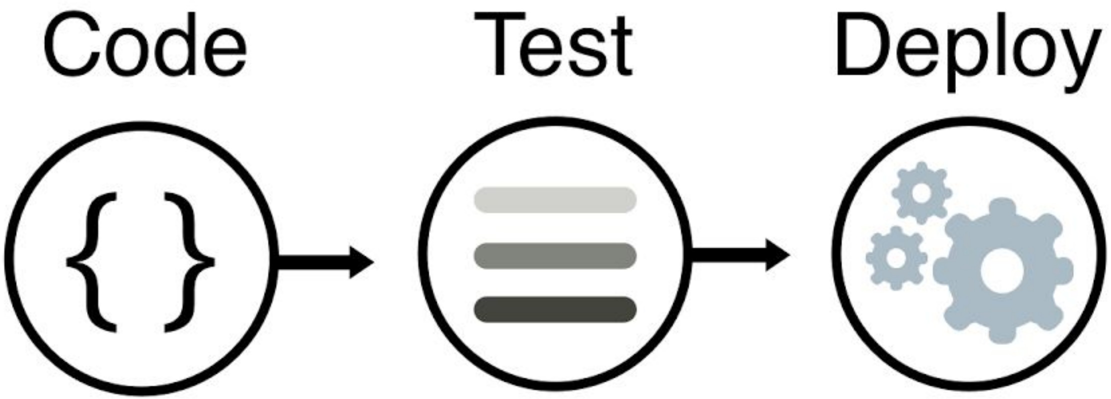
DOCKER RUN

```
docker run -d --name=redis redis

docker run -d --name=db postgres:9.4

docker run -d --name=vote -p 5000:80  voting-app

docker run -d --name=result -p 5001:80  result-app

docker run -d --name=worker  worker
```

Continous Integration/ Continous Deployment/ Continous Delivery

CI/CD

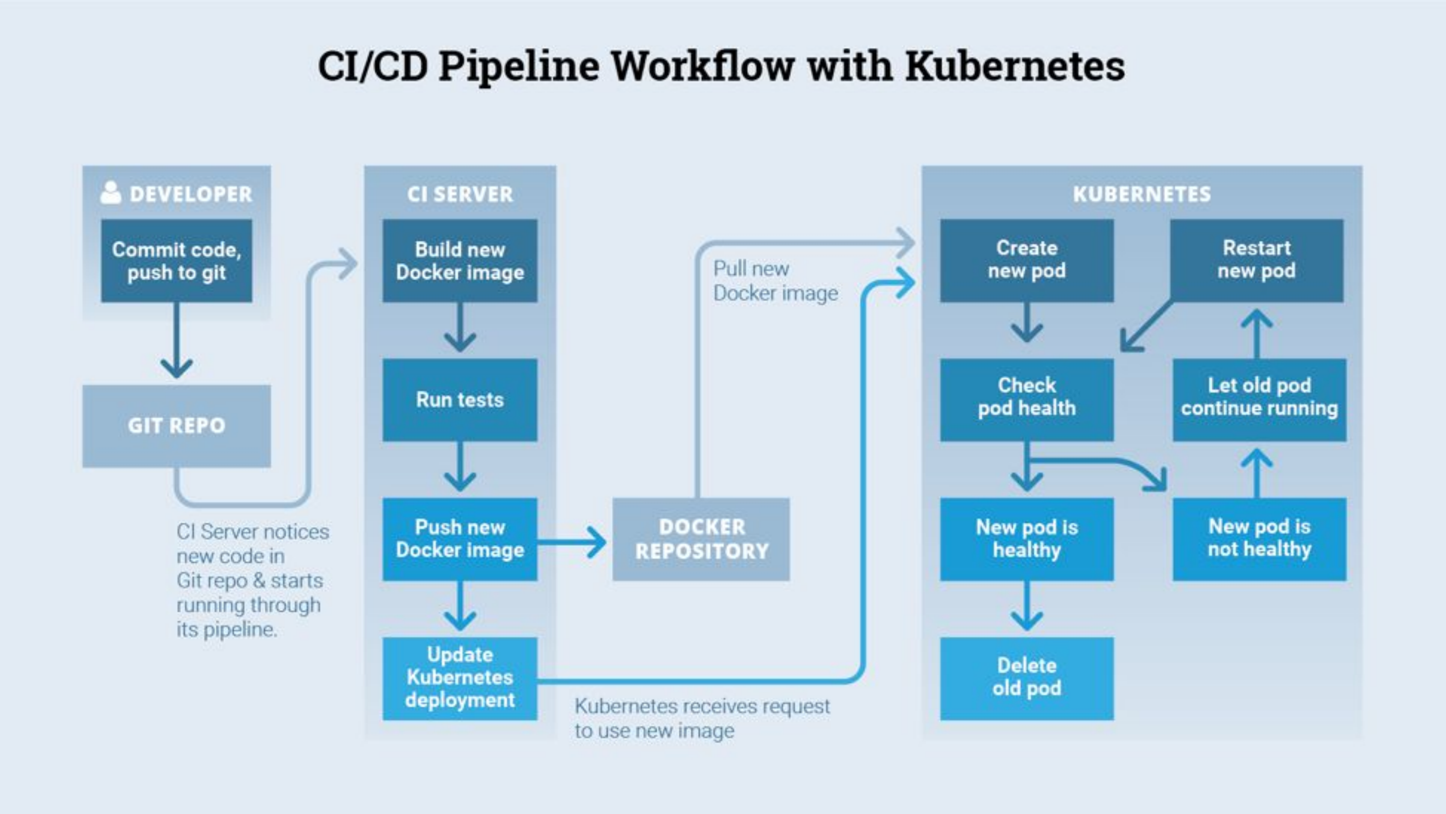
CI/CD (Continuous Integration/Continuous Delivery) is a methodology that streamlines software development through collaboration and automation and is a critical component of implementing DevOps.



CI/CD

- Jenkins
- TeamCity
- GitLab CI/CD
- CircleCI
- Travis CI
- Drone CI (Special Mention)

CI/CD



CI/CD Pipeline Workflow with Kubernetes
DEVELOPER
Commit code, push to git
GIT REPO
CI Server notices new code in Git repo & starts running through its pipeline.
CI SERVER
Build new Docker image
Run tests
Push new Docker image
Update Kubernetes deployment
DOCKER REPOSITORY
Pull new Docker image
Kubernetes receives request to use new image
KUBERNETES
Create new pod
Restart new pod
Check pod health
Let old pod continue running
New pod is healthy
New pod is not healthy
Delete old pod

CI/CD Tools

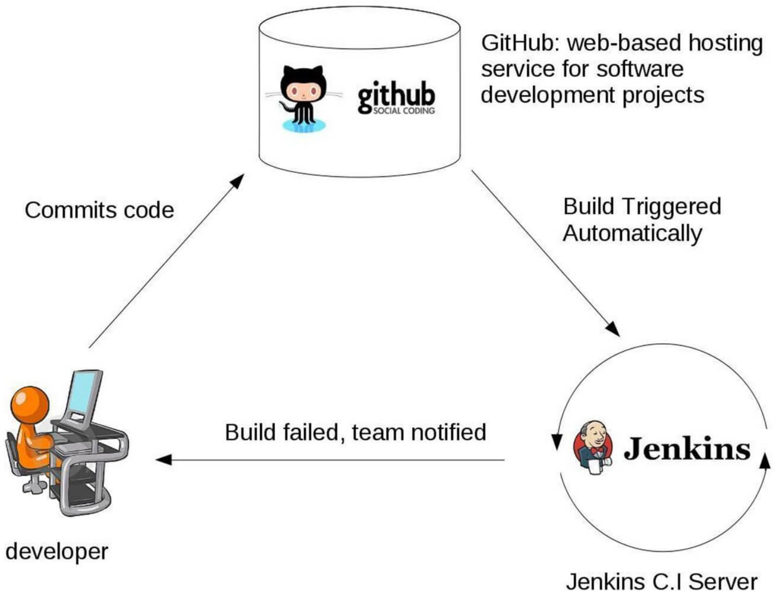
Package Managers - Helm

- Like yum/apt/homebrew for k8s
- Helm is a tool that streamlines installing and managing Kubernetes applications.
- Helm has two parts: a client (helm) and a server (iller) Tiller runs inside of your Kubernetes cluster, and manages releases (installations) of your charts.
- Helm Charts help you define, install, and upgrade even the most complex Kubernetes application.

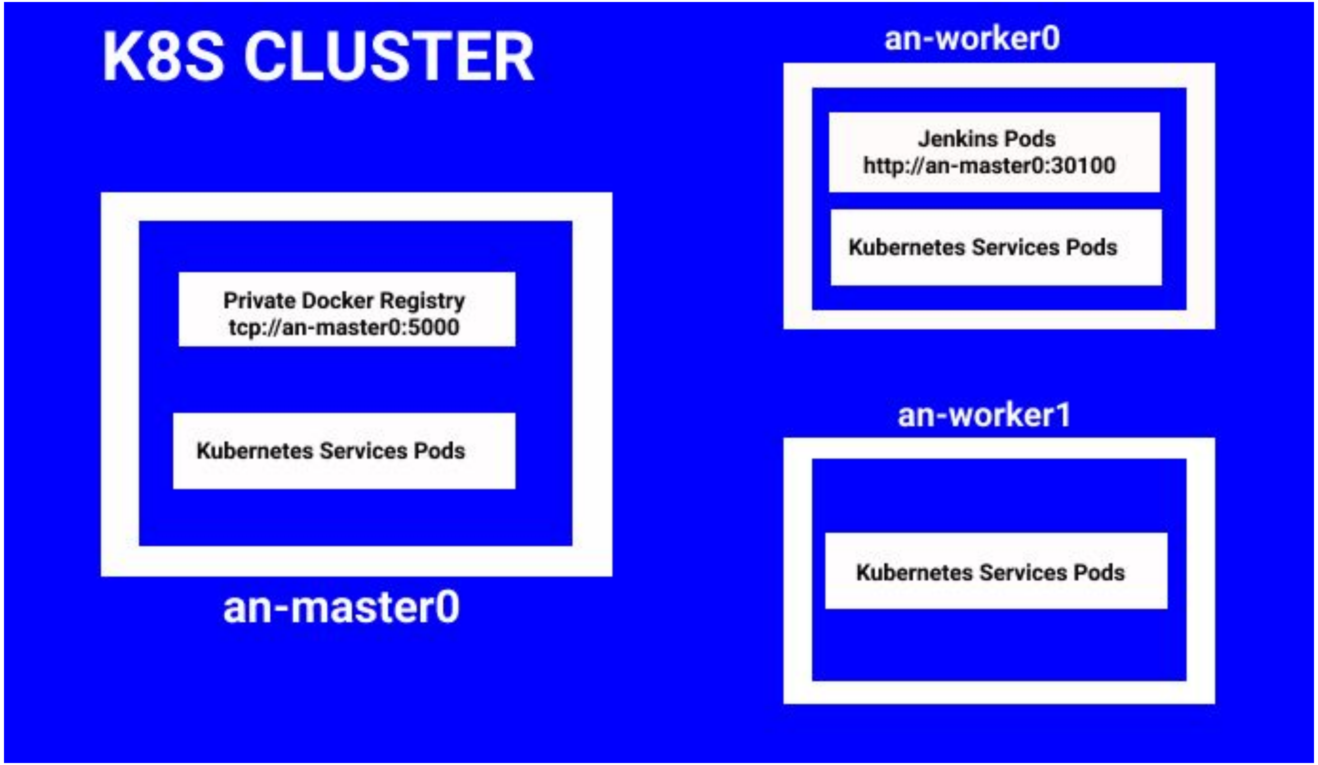
CI/CD Tools - Jenkins

- Continuous Intregration Server
- Originally develop as The Hudson project by Sun Microsystems
- Reliably build, test, and deploy their software.
- Free

CI/CD Tools - Jenkins



CI/CD Tools - Jenkins (DEMO)



CI/CD Tools - Jenkins (DEMO)

Step :

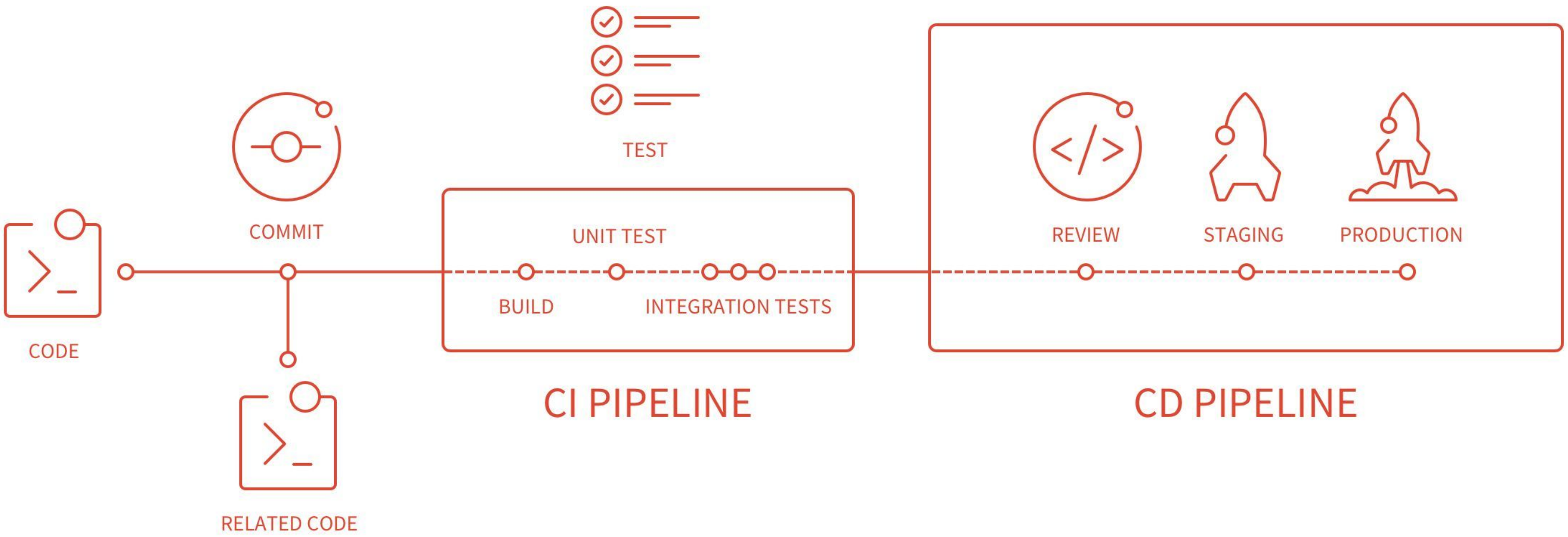
- Set up a Jenkins environment on Kubernetes.
- Configure a CI/CD Jenkins pipeline.
- Dockerize App
- Push images to docker registry.
- Pull new images from registry.
- Deploy the app on Kubernetes.
- See this link for more detail : https://s.id/8MnoL

CI/CD Tools - Gitlab

GitLab

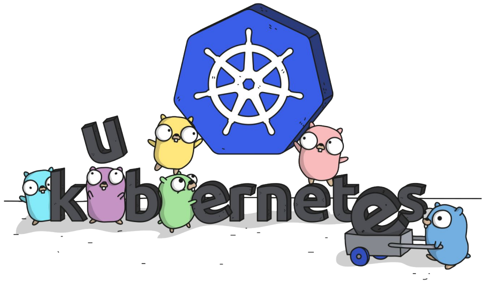
- GitLab is a single application for the entire DevOps lifecycle that allows teams to work together better and bring more value to your customers, faster.
- GitLab does this by shortening your devops cycle time, bridging silos and stages, and taking work out of your hands. It also means a united workflow that reduces friction from traditionally separate activities like application security testing.

CI/CD Tools - Gitlab



CI/CD Tools - Gitlab (DEMO)

- Create a Kubernetes Cluster
- Deploy gitlab in Kubernetes Cluster
- Create the project in Gitlab
- Create CI/CD Pipeline
- Deploy!
- See this link for more detail : https://s.id/8Mnuz



Ask Me Question?